Fluid Mechanics & Hydraulic Machinery Lab (23A03402P)

II Year B.Tech ME, II Semester | JNTUA R23

Course code: 23A03402P

Credits: L = 0, T = 0, P = 3, C = 1.5

Course Objective: To impart practical exposure on the performance evaluation methods of various flow measuring equipment, hydraulic turbines and pumps.

Course Outcomes

CO	Statement	Blooms
CO1	Demonstrate the devices used for measuring flow.	L3
CO2	Compute major losses in pipes.	L5
CO3	Illustrate the operating parameters of turbines.	L2
CO4	Explain the working of different types of pumps.	L2
CO5	Explain the devices used for measuring flow.	L2

Related COs shown for each experiment are suggested by topic.

Each experiment lists: Related CO, Aim, Apparatus, INPUTS (what you set / measure), Procedure, OUTPUTS (what you calculate / get).

List of Experiments

1. Impact of jets on vanes
2. Performance test on Pelton wheel
3. Performance test on Francis turbine
4. Performance test on Kaplan turbine
5. Performance test on single stage centrifugal pump
6. Performance test on multi stage centrifugal pump
7. Performance test on reciprocating pump
8. Calibration of venturimeter
9. Calibration of orifice meter
10. Determination of friction factor for a given pipeline
11. Determination of loss of head due to sudden contraction in a pipeline
12. Turbine flow meter

Virtual lab: 7 simulations (at the end)

1. Impact of jets on vanes

Related CO: CO3

Aim: To find the force exerted by a water jet on vanes and the coefficient of impact.

Apparatus: Jet apparatus with nozzle, vanes (flat, hemispherical, inclined), weights, collecting tank, stop watch.

INPUTS: Nozzle diameter d; vane type; weights placed on the pan (balancing); water collected in the tank: volume or height in time t.

Procedure:

1) Fix a vane and start the pump.
2) Balance the vane position by adding weights.
3) Measure discharge using the collecting tank and stop watch.
4) Repeat for other vanes and flow rates.

OUTPUTS: Actual force Fa, theoretical force Ft and coefficient of impact K = Fa/Ft for each vane.

Formulae: Q = Volume/t; V = Q/a; flat plate Ft = rho a V^2; hemispherical Ft = 2 rho a V^2

2. Performance test on Pelton wheel

Related CO: CO3

Aim: To find the efficiency of a Pelton wheel at different loads and plot characteristic curves.

Apparatus: Pelton turbine test rig, pump, spear valve, rope brake dynamometer, tachometer, pressure gauge, collecting tank.

INPUTS: Head H (pressure gauge); speed N (rpm); brake loads W1 and W2; drum radius R; discharge Q (tank and stop watch); nozzle opening.

Procedure:

1) Start the pump and set the spear valve.
2) Note the pressure gauge reading and discharge.
3) Vary the brake load and note N, W1, W2.
4) Repeat at different loads.

OUTPUTS: Brake power, water power, overall efficiency; curves: efficiency vs speed and power vs speed.

Formulae: BP = 2 pi N T / 60 (T = (W1 - W2) R); WP = rho g Q H; efficiency = BP/WP x 100

3. Performance test on Francis turbine

Related CO: CO3

Aim: To find the efficiency of a Francis turbine and draw characteristic curves.

Apparatus: Francis turbine test rig, rope brake, tachometer, pressure and vacuum gauges, venturi or tank for flow measurement.

INPUTS: Pressure gauge reading, vacuum gauge reading, speed N, brake loads W1 and W2, discharge Q, guide vane opening.

Procedure:

1) Start the pump, keep the guide vane open.
2) Note the pressure, vacuum, flow and speed.
3) Load the brake and note readings again.
4) Repeat at several loads.

OUTPUTS: Head, brake power, water power, efficiency; graphs of efficiency, power and discharge vs speed.

Formulae: H = pressure head + vacuum head; BP = 2 pi N T/60; WP = rho g Q H; efficiency = BP/WP

4. Performance test on Kaplan turbine

Related CO: CO3

Aim: To find the efficiency of a Kaplan turbine at different gate openings.

Apparatus: Kaplan turbine test set up, brake drum, tachometer, pressure and vacuum gauges, discharge measuring unit.

INPUTS: Gate (guide vane) opening; runner blade angle; head H; speed N; brake loads; discharge Q.

Procedure:

1) Start the pump and set the guide vane.
2) Take readings at no load and then add load.
3) Repeat at another gate opening.

OUTPUTS: Brake power, water power, efficiency; characteristic curves for constant head and constant gate opening.

Formulae: BP = 2 pi N T/60; WP = rho g Q H; eta = BP/WP

5. Performance test on single stage centrifugal pump

Related CO: CO4

Aim: To find the efficiency of a single stage centrifugal pump and draw the characteristics.

Apparatus: Centrifugal pump with motor, suction and delivery pressure gauges, delivery valve, energy meter or wattmeter, collecting tank.

INPUTS: Suction (vacuum) head Hs; delivery pressure head Hd; discharge Q (tank rise in time t); energy meter pulses or power input; speed N.

Procedure:

1) Prime the pump and start the motor.
2) Keep the delivery valve fully open then reduce it step by step.
3) Note gauge readings, discharge and energy meter time for each setting.

OUTPUTS: Total head, input power, output power, overall efficiency; curves: H, power and efficiency vs Q.

Formulae: H = Hd + Hs; output power = rho g Q H; input power from energy meter; eta = output / input x 100

6. Performance test on multi stage centrifugal pump

Related CO: CO4

Aim: To find the performance of a multi stage centrifugal pump (stages in series or parallel).

Apparatus: Multi stage pump with motor, gauges, valves, tank, energy meter.

INPUTS: Number of stages in use; suction vacuum; delivery pressure; discharge Q; power input.

Procedure:

1) Start the pump with all valves set for the chosen stage arrangement.
2) Vary the delivery valve and record readings.
3) Repeat with another arrangement.

OUTPUTS: Head, discharge, input and output power and efficiency; head vs Q and efficiency vs Q graphs.

Formulae: Same as single stage: output = rho g Q H; eta = output/input

7. Performance test on reciprocating pump

Related CO: CO4

Aim: To find the discharge, slip and efficiency of a reciprocating pump.

Apparatus: Reciprocating pump with motor, suction vacuum and delivery pressure gauges, tank, energy meter, tachometer.

INPUTS: Cylinder bore D, stroke L, speed N, pressure gauge reading, vacuum reading, actual discharge Q (tank), power input.

Procedure:

1) Start the pump and fix a delivery pressure.
2) Note speed, discharge and energy meter time.
3) Repeat for different delivery pressures.

OUTPUTS: Theoretical discharge, % slip, coefficient of discharge, total head, overall efficiency.

Formulae: Qth = A L N / 60; slip % = (Qth - Qa)/Qth x 100; Cd = Qa/Qth; eta = rho g Qa H / input power

8. Calibration of venturimeter

Related CO: CO1, CO5

Aim: To find the coefficient of discharge of a venturimeter.

Apparatus: Venturimeter in a pipe line, U-tube mercury manometer, collecting tank, stop watch.

INPUTS: Inlet and throat diameters d1, d2; manometer difference h_m; discharge by tank (volume/time); several flow rates.

Procedure:

1) Run water through the pipe at a set flow.
2) Note the manometer difference.
3) Measure the discharge using the tank.
4) Repeat for different flow rates.

OUTPUTS: Cd for each flow rate and its mean value; plot of Qact vs sqrt(h).

Formulae: h = h_m (Sm/Sw - 1); Qth = a1 a2 sqrt(2 g h) / sqrt(a1^2 - a2^2); Cd = Qact / Qth

9. Calibration of orifice meter

Related CO: CO1, CO5

Aim: To find the coefficient of discharge of an orifice meter.

Apparatus: Orifice meter in a pipe line, mercury U-tube manometer, tank and stop watch.

INPUTS: Pipe diameter D, orifice diameter d0, manometer reading h_m, discharge Q (volume/time).

Procedure:

1) Set a flow rate and note the manometer reading.
2) Measure the actual discharge.
3) Repeat for several flows.

OUTPUTS: Cd of the orifice meter (about 0.6 to 0.65) and the calibration curve.

Formulae: Qth = a0 a1 sqrt(2 g h) / sqrt(a1^2 - a0^2); Cd = Qact/Qth

10. Determination of friction factor for a given pipeline

Related CO: CO2

Aim: To find the friction factor of a given pipe.

Apparatus: Pipe test rig with different pipe sizes, differential manometer, flow control valve, tank and stop watch.

INPUTS: Pipe diameter d, length L between tapings, manometer reading (head loss hf), discharge Q.

Procedure:

1) Start flow and control with the valve.
2) Note the manometer reading and discharge.
3) Repeat for various flow rates and pipes.

OUTPUTS: Velocity, head loss and Darcy friction factor f for each trial; graph f vs Reynolds number.

Formulae: v = Q/A; hf = h_m (Sm/Sw - 1); f = 2 g d hf / (L v^2); Re = v d / nu

11. Determination of loss of head due to sudden contraction in a pipeline

Related CO: CO2

Aim: To find the head loss and loss coefficient due to a sudden contraction.

Apparatus: Pipe line with sudden contraction (large to small pipe), piezometers or manometer, tank, stop watch.

INPUTS: Diameters D1 and D2; piezometer heads before and after the contraction; discharge Q.

Procedure:

1) Run water at a set flow.
2) Read heads h1 and h2 at the two taping points.
3) Find the discharge.
4) Repeat for different flow rates.

OUTPUTS: Head loss hL and loss coefficient K for the contraction.

Formulae: V1 = Q/A1, V2 = Q/A2; hL = (h1 - h2) + (V1^2 - V2^2)/2g; K = hL / (V2^2/2g)

12. Turbine flow meter

Related CO: CO1, CO5

Aim: To calibrate a turbine flow meter and study its performance.

Apparatus: Turbine flow meter with digital display, pump, control valve, collecting tank, stop watch.

INPUTS: Flow rate set by the valve; pulse count or rpm of the turbine meter; actual discharge from the tank.

Procedure:

1) Start the pump and set the flow.
2) Note the digital meter reading and the actual discharge.
3) Repeat at different flow rates.

OUTPUTS: Calibration curve (meter reading vs actual discharge), meter factor (K-factor), and percentage error.

Formulae: Actual Q = Volume / t; % error = (Q meter - Q actual)/Q actual x 100

Virtual Lab

1. Flow patterns in a pipe and Reynolds number

Link: https://me.iitp.ac.in/Virtual-Fluid-Laboratory/reynolds/introduction.html

INPUTS: Flow rate (velocity), pipe diameter, dye injection.

OUTPUTS: Laminar, transition or turbulent pattern; Reynolds number Re = v d / nu.

2. Total energy at different points of a venturi meter

Link: https://me.iitp.ac.in/Virtual-Fluid-Laboratory/bernoulli/introduction.html

INPUTS: Flow rate and pipe sections (point readings of pressure head).

OUTPUTS: Pressure head, velocity head, datum head and total energy at each point (Bernoulli).

3. Velocity at the centre of a tube (Pitot tube)

Link: https://me.iitp.ac.in/Virtual-Fluid-Laboratory/pitot/introduction.html

INPUTS: Different flow rates; manometer difference of the pitot tube.

OUTPUTS: Centre-line velocity for each flow rate; v = Cv sqrt(2 g h).

4. Hydrostatic force on a plane surface

Link: https://me.iitp.ac.in/Virtual-Fluid-Laboratory/cop/introduction.html

INPUTS: Water depth; partially or fully submerged condition; weights on the balance.

OUTPUTS: Hydrostatic force F = rho g A h_bar and centre of pressure.

5. Discharge coefficient of a triangular notch

Link: https://me.iitp.ac.in/Virtual-Fluid-Laboratory/notch/introduction.html

INPUTS: Head over the notch H; notch angle; actual discharge.

OUTPUTS: Cd of the notch; Q = (8/15) Cd sqrt(2g) tan(theta/2) H^(5/2).

6. Coefficient of impact of jet on vanes

Link: https://fm-nitk.vlabs.ac.in/exp/impact-of-jet

INPUTS: Jet velocity, nozzle diameter, vane type, weights/force.

OUTPUTS: Actual and theoretical force; coefficient of impact K.

7. Friction in pipes

Link: https://fm-nitk.vlabs.ac.in/exp/friction-in-pipes/index.html

INPUTS: Pipe diameter, length, flow rate, manometer reading.

OUTPUTS: Head loss hf and friction factor f; graph f vs Re.